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BUSBAR AND BATTERY MODULE

Abstract

The application provides a busbar and a battery module, which relate to the technical field of battery. The busbar includes a bar part and a plurality of conductors arranged at intervals. Each of the plurality of conductors includes a fuse part and a conducting part. The fuse part has a melting point lower than a melting point of the bus bar and lower than a melting point of the conducting part. Two ends of the fuse part are respectively connected to the conducting part and the bar part. The conducting part is configured to be connected to a positive electrode or a negative electrode of a battery cell.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to and the benefit of Chinese Patent Application No. 202420407843.2, filed on Mar. 1, 2024, and PCT Application No. PCT/CN2024/097050, filed on Jun. 3, 2024, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present application relates to the technical field of battery, and in particular to a bus and a battery module.

BACKGROUND

[0003] In the related technologies, as the application of power batteries becomes more and more extensive, their safety issues have received more and more attention. Battery short circuit is an important factor causing safety issues. At present, the design requirements of battery short circuit protection are mainly met by designing a fuse structure on the busbar, so that when the battery is short-circuited or the temperature is abnormal, the fuse structure is melted, thereby disconnecting the battery circuit.

[0004] Usually, there are multiple busbars configured to connect multiple battery cells in series in sequence, that is, one side of the busbar is a positive electrode conducting part electrically connected to the positive electrode of one battery cell, and the other side is a negative electrode conducting part electrically connected to the negative electrode of another battery cell, so as to connect adjacent battery cells in series.

[0005] Although the fuse structure is arranged between the positive electrode conducting part and the negative electrode conducting part to meet the fusing requirement of the busbar, it will reduce the current carrying capacity of the busbar, thereby causing an increase in the internal resistance of the battery module.

SUMMARY

[0006] In a first aspect, an embodiment of the present application provides a busbar, including:

[0007] a bar part; [0008] a plurality of conductors arranged at intervals, each of the plurality of conductors including a fuse part and a conducting part, wherein the fuse part has a melting point lower than a melting point of the bar part and lower than a melting point of the conducting part, two ends of the fuse part are respectively connected to the conducting part and the bar part, and the conducting part is configured to be connected to a positive electrode or a negative electrode of a battery cell.

[0009] In a second aspect, an embodiment of the present application provides a battery module, including the above busbar.

[0010] According to the present application, each fuse part is arranged between the bar part and the corresponding conducting part, so that the current gathered by the busbar does not pass through the fuse parts. Therefore, the fuse parts can meet the fusing requirement of the busbar without affecting the amount of the current that the busbar can carry, thereby improving the current carrying capacity of the busbar and reducing the internal resistance of the battery module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a schematic diagram of the structure of a busbar according to an embodiment of the present application;

[0012] FIG. **2** is a schematic diagram of a partial structure of a busbar according to an embodiment of the present application;

[0013] FIG. **3** is an enlarged view of Portion A in FIG. **1**;

[0014] FIG. **4** is an exploded view of a busbar according to an embodiment of the present application.

[0015] List of Reference Numerals [0016] **001**, busbar; [0017] **011**, conductor; **111**, conducting part; **1111**, first through hole; **112**, fuse part; **1121**, fuse hole; [0018] **012**, bar part; **121**, first elongated member; **122**, second elongated member; **123**, positioning hole; **124**, second through hole.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0019] The terms “first” and “second” are used for descriptive purposes only and should not be understood as indicating or implying relative importance or implicitly indicating the number of the indicated technical features. Thus, a feature defined as “first” or “second” may explicitly or implicitly include one or more of the features. In the description of this application, the meaning of “plurality” is two or more, unless otherwise clearly and specifically defined.

[0020] In the description of this application, it should be noted that, unless otherwise clearly specified and limited, the terms “installed”, “connected” and “coupled” should be understood in a broad sense, for example, it can be a fixed connection, a detachable connection, or an integral connection; it can be a mechanical connection, an electrical connection, or mutual communication; it can be a direct connection, or an indirect connection through an intermediate medium, it can be the internal connection of two elements or the interaction relationship between two elements. For ordinary technicians in this field, the specific meanings of the above terms in this application can be understood according to specific circumstances.

[0021] In the description of the embodiments of the present application, words such as “example” or “for example” are used to indicate examples, illustrations or descriptions. Any embodiment or design described as “example” or “for example” in the embodiments of the present application is not to be interpreted as being more preferred or having more advantages than another embodiment or design. The use of words such as “example” or “for example” is intended to present relative concepts in a clear manner.

[0022] Please refer to FIG. **1**, which is a schematic diagram of the structure of a busbar **001** according to an embodiment of the present application. An embodiment of the present application provides a busbar **001**. The busbar **001** includes a bar part **012** and a plurality of conductors **011** arranged at intervals. The conductors **011** each include a fuse part **112** and a conducting part **111**. The melting point of each fuse part **112** is lower than the melting points of the bar part **012** and the corresponding conducting part **111**. Each fuse part **112** has two ends respectively connected to the corresponding conducting part **111** and the bar part **012**. The conducting parts **111** each are configured to be connected to the positive electrode or the negative electrode of a battery cell.

[0023] It can be understood that the plurality of conductors **011** are arranged at intervals along the length direction of the bar part **012**. A plurality of conducting parts **111** are connected to the same bar part **012** and are connected to electrodes of the same polarity of a plurality of battery cells, thereby connecting the positive or negative electrodes of the plurality of battery cells in parallel. Accordingly, the battery module is provided with at least two busbars, one of which is configured to connect the positive electrodes of the plurality of battery cells in parallel, and the other is configured to connect the negative electrodes of the plurality of battery cells in parallel. The busbar **001** according to this embodiment may be used to connect the positive electrodes of the battery cells, and a conventional busbar may be used to connect the negative electrodes of the battery cells.

[0024] Exemplarily, the melting point of each fuse part **112** is less than 1000° C., and the melting points of the bar part **012** and each conducting part **111** are greater than 1000° C. When the current is abnormal and causes a rise in temperature to a certain degree, the fuse parts **112** melts themselves to cut off the current, thereby disconnecting the battery circuit.

[0025] There are many ways to fuse the fuse parts **112** so as to disconnect the bar part **012** from the conducting parts **111**. For example, in a first way, the fuse parts **112** may be formed of a material having a melting point lower than the melting point of the material of the bar part **012** and also lower than the melting point of the material of the conducting parts **111**, so that the fuse parts **112** can be fused earlier than other components. For another example, in a second way, the cross section of each fuse part **112** perpendicular to the current flowing direction through the fuse part **112** may be structured to be smaller than the cross section of the corresponding conducting part **111** perpendicular to the current flowing direction through the conducting part **111** and also smaller than the cross section of the bar part **012** perpendicular to the current flowing direction through the bar part **012**, so that the fuse parts **112** can be fused earlier than other components.

[0026] In this embodiment, each fuse part **112** is arranged between the bar part **012** and the corresponding conducting part **111**, so that the current gathered by the busbar **001** does not pass through the fuse parts **112**. Therefore, the fuse parts **112** can meet the fusing requirement of the busbar **001** without affecting the amount of the current that the busbar **001** can carry, thereby improving the current carrying capacity of the busbar **001** and reducing the internal resistance of the battery module.

[0027] Please refer to FIG. 2, which is a schematic diagram of a partial structure of a busbar **001** according to an embodiment of the present application. In an embodiment, each fuse part **112** extends from the bar part **012** to the corresponding conducting part **111**, and the extension direction of the fuse part **112** forms an acute angle or an obtuse angle with the extension direction of the bar part **012**, so that the two ends of the fuse part **112** are staggered along the length direction of the bar part **012**.

[0028] In this embodiment, the extension direction of each fuse part **112** forms an acute angle or an obtuse angle with the extension direction of the bar part **012**, so that the two ends of the fuse part **112** are staggered along the length direction of the bar part **012**. In this way, the size of each fuse part **112** in the length direction of the bar part **012** can be increased while the spacing between the conducting parts **111** and the bar part **012** is limited, thereby making the size of the fuse part **112** meet the fusing requirements, and finally improving the reliability of the busbar **001**.

[0029] In an embodiment, the fuse parts **112** are configured to be fused earlier than other components in the above mentioned second way. Specifically, the cross section of each fuse part **112** perpendicular to the current flowing direction through the fuse part **112** is smaller than the cross section of the corresponding conducting part **111** perpendicular to the current flowing direction through the conducting part **111** and is also smaller than the cross section of the bar part **012** perpendicular to the current flowing direction through the bar part **012**.

[0030] Exemplarily, each fuse part **112** is integrally formed with the corresponding conducting part **111**.

[0031] In this embodiment, the fusing of the fuse parts **112** is achieved earlier than other components, which depends on the size relation between the cross sections of various components where the current flows. Therefore, the fuse parts **112** and the conducting parts **111** can be formed of the same material, so that each fuse part **112** can be integrally formed with the corresponding conducting part **111**, thereby improving the efficiency in formation of the conductors **011** and reducing the manufacturing difficulty of the busbar **001**.

[0032] In an embodiment, a fuse hole **1121** or a fuse slot is defined on each fuse part **112**.

[0033] In this embodiment, by defining a fuse hole **1121** or a fuse slot on each fuse parts **112**, the cross section of the fuse part **112** perpendicular to the current flowing direction through the fuse part **112** can be made smaller, and the fuse part **112** has a larger surface area, thereby improving the

fusing reliability of the fuse part **112**.

[0034] Specifically, in the case that a fuse hole **1121** is defined on a fuse part **112**, as shown in FIG. 2, the fuse hole **1121** is an elongated kidney-shaped hole, and the extension direction of walls of the fuse hole **1121** is consistent with the extension direction of the fuse part **112**. In the case that a fuse slot is defined on a fuse part **112**, the fuse slot has an elongated shape, and the extension direction of walls of the fuse slot is consistent with the extension direction of the fuse part **112**. In this way, the length of the structure of the fuse part **112** for fusing can be increased, so that fusing at any position of the walls of the fuse hole **1121** or at any position of the walls of the fuse slot can meet the fusing requirements, thereby improving the fusing reliability of the fuse part **112**.

[0035] Please refer to FIG. 2. In an embodiment, a fuse part **112** has an arc-shaped structure, and two ends of the arc-shaped structure are respectively connected to the corresponding conducting part **111** and the bar part **012**.

[0036] Based on the above arrangement according to this embodiment, the fuse part **112** can be longer in size when the distance between the two ends of the fuse part **112** is fixed, thereby further improving the reliability of the busbar **001**.

[0037] Please refer to FIG. 1, in an embodiment, the cross section of the bar part **012** perpendicular to the current flowing direction through the bar part **012** is larger than the cross section of each conductor **011** perpendicular to the current flowing direction through the conductor **011**.

[0038] It can be understood that the current passing through bar part **012** is greater than the current passing through each conductor **011**. Therefore, in order to meet the overcurrent requirement of bar part **012**, the cross section of bar part **012** perpendicular to the current flowing direction through the bar part **012** is structured to be larger than the cross section of each conductor **011** perpendicular to the current flowing direction through the conductor **011**, so as to improve the reliability of busbar **001**.

[0039] Exemplarily, thirteen conductors **011** are provided on a side of the bar part **012**.

[0040] In combination with the above-mentioned embodiments, the bending directions of all the fuse parts **112** of the conductors **011** may be consistent, or the bending directions of some of the fuse parts **112** of the conductors **011** may be consistent and the bending directions of the others of the fuse parts **112** of the conductors **011** are opposite to the bending directions of the some of the fuse parts **112** of the conductors **011**, as shown in FIG. 1. Specifically, the bending directions of the fuse parts **112** depends on the arrangements of different battery modules.

[0041] Please refer to FIG. 3, which is an enlarged view of Portion A in FIG. 1. In an embodiment, the thickness of the bar part **012** is greater than the thickness of each conductor **011**.

[0042] Exemplarily, the thickness of the bar part **012** is 2-2.6 mm, and the thickness of each conductor **011** is 0.2-0.4 mm. It is understood that the thickness of the bar part **012** includes but is not limited to 2 mm, 2.1 mm, 2.2 mm, 2.4 mm, 2.5 mm, and 2.6 mm. The thickness of each conductor **011** includes but is not limited to 0.2 mm, 0.21 mm, 0.22 mm, 0.3 mm, 0.35 mm, and 0.4 mm.

[0043] In this embodiment, a larger cross section of the bar part **012** perpendicular to the current flowing direction through the bar part **012** is achieved by increasing the thickness of the bar part **012**, which can reduce the surface area of the bar part **012** in the battery module, thereby reducing the difficulty of arranging the busbar **001** and improving the installation efficiency of the busbar **001**.

[0044] In an embodiment, the bar part **012** is integrally formed with the conductors **011**.

[0045] Based on the above arrangement according to this embodiment, the reliability of the connection between the bar part **012** and the conductors **011** can be improved, thereby improving the reliability of the busbar **001**.

[0046] Please refer to FIG. 4, which is an exploded view of a busbar **001** according to an embodiment of the present application. In an embodiment, the bar part **012** includes a first elongated member **121** and a second elongated member **122** stacked in sequence along the

thickness direction of the bar part **012**, the second elongated member **122** is integrally formed with the conductors **011**, and the thickness of the second elongated member **122** is consistent with the thickness of the conductors **011**.

[0047] As can be understood, in the case that the bar part **012** is integrally formed with the conductors **011**, the thickness difference between the bar part **012** and the conductors **011** increases the difficulty in formation. In view of this, in this embodiment, the bar part **012** is structured as a combination of first elongated member **121** and a second elongated member **122**, and the thickness of the second elongated member **122** is consistent with the thickness of the conductors **011**. The first elongated member **121** and the combination of the second elongated member **122** and the conductors **011** are formed respectively in such a way that the thickness of the first elongated member **121** is same as the overall thickness of the combination of the second elongated member **122** and the conductors **011**. Therefore, the difficulty in forming the busbar **001** can be reduced.

[0048] Furthermore, the first elongated member **121** is welded with the second elongated member **122**. Specifically, the first elongated member **121** and the second elongated member **122** are combined into one by laser welding.

[0049] In an embodiment, the melting point of the first elongated member **121** is greater than the melting point of the conductors **011**.

[0050] Specifically, the material of the first elongated member **121** may be copper, and the material of the second elongated member **122** may be aluminum.

[0051] It can be understood that the second elongated member **122** is integrally formed with the conductors **011**. Therefore, the second elongated member **122** and the conductors **011** are made of the same material. Accordingly, the melting point of the first elongated member **121** is also greater than the melting point of the conductors **011**.

[0052] Based on the above arrangement according to this embodiment, the current carrying capacity of the bar part **012** can be improved, and the bar part **012** can be prevented from being fused, thereby improving the reliability of the busbar **001**.

[0053] In an embodiment, the resistivity of the first elongated member **121** is lower than the resistivity of the second elongated member **122**.

[0054] It can be understood that the resistance of the first elongated member **121** is lower than the resistance of the second elongated member **122**.

[0055] Specifically, the material of the first elongated member **121** may be copper, and the material of the second elongated member **122** may be aluminum.

[0056] Based on the above arrangement according to this embodiment, the conducting capacity of the busbar **001** can be improved and the material cost of the busbar **001** can be controlled.

[0057] In an embodiment, the busbar **001** is applied to a battery module. The battery module includes battery cells. The battery cells each have poles. A first through hole **1111** is defined on each conducting part **111**, and the first through hole **1111** is configured to be aligned with one of the poles.

[0058] It can be understood that a reference ring is provided on the end face of each pole, and the reference ring is arranged coaxially with the pole. When the first through hole **1111** of a conducting part **111** shows therefrom the reference ring of a corresponding pole completely, it can be determined that the conducting part **111** is aligned with the pole.

[0059] In this embodiment, by providing the first through holes **1111**, it is convenient for a person to quickly determine whether the busbar **001** is installed in place, thereby improving the convenience of the installation of the busbar **001**.

[0060] According to the embodiment of the present application, the busbar **001** can be installed in the following way. A positioning hole **123** and a second through hole **124** are defined on the bar part **012**. The positioning hole **123** is configured to match with the positioning column on the insulating bracket of the battery module. The second through hole **124** is configured to match with the screw. The threaded end of the screw extends through the second through hole **124** and then is

threadedly connected to the insulating bracket, thereby fixing the busbar **001** on the insulating bracket. Then, the conducting parts **111** are fixedly connected to the electrodes of the battery, such as the positive electrodes, by laser welding.

[0061] Correspondingly, an embodiment of the present application further provides a battery module, which includes the aforementioned busbar.

[0062] It can be understood that the battery module also includes battery cells, a cell bracket, etc. The battery cells are installed on the cell bracket. A collection bracket is fixedly connected to ends of the battery cells, and the busbar is arranged on the collection bracket and is electrically connected to the electrodes of the battery cells.

[0063] In this embodiment, the aforementioned busbar allows an increased size of each fuse part **112** in the length direction of the bar part **012** in the case that the distance between the conducting parts **111** and the bar part **012** is limited, so that the size of the fuse part **112** meets the fusing requirements, and finally the reliability of the battery module can be improved.

Claims

1. A busbar, comprising: a bar part; a plurality of conductors arranged at intervals, each of the plurality of conductors comprising a fuse part and a conducting part, wherein the fuse part has a melting point lower than a melting point of the bar part and lower than a melting point of the conducting part, two ends of the fuse part are respectively connected to the conducting part and the bar part, and the conducting part is configured to be connected to a positive electrode or a negative electrode of a battery cell.
2. The busbar according to claim 1, wherein the fuse part extends from the bar part toward the conducting part, and an extending direction of the fuse part forms an acute angle or an obtuse angle with an extending direction of the bar part.
3. The busbar according to claim 1, wherein a cross section of the fuse part perpendicular to a current flowing direction through the fuse part is smaller than a cross section of the conducting part perpendicular to a current flowing direction through the conducting part and is smaller than a cross section of the bar part perpendicular to the current flowing direction through the bar part.
4. The busbar according to claim 3, wherein a fuse hole is defined on the fuse part.
5. The busbar according to claim 3, wherein a fuse slot is defined on the fuse part.
6. The busbar according to claim 4, wherein the fuse hole has an elongated shape, and an extension direction of a wall of the fuse hole is consistent with an extension direction of the fuse part.
7. The busbar according to claim 5, wherein the fuse slot has an elongated shape, and an extension direction of a wall of the fuse slot is consistent with an extension direction of the fuse part.
8. The busbar according to claim 1, wherein the fuse part has an arc-shaped structure with two ends thereof respectively connected to the conducting part and the bar part.
9. The busbar according to claim 1, wherein a cross section of the bar part perpendicular to a current flowing direction through the bar part is larger than a cross section of each of the plurality of conductors perpendicular to a current flowing direction through the conductor.
10. The busbar according to claim 9, wherein a thickness of the bar part is greater than a thickness of each of the plurality of conductors.
11. The busbar according to claim 10, wherein the bar part comprises a first elongated member and a second elongated member stacked in sequence along a thickness direction of the bar part, and wherein the second elongated member is integrally formed with the plurality of conductors, and a thickness of the second elongated member is consistent with the thickness of each of the plurality of conductors.
12. The busbar according to claim 11, wherein a resistivity of the first elongated member is lower than a resistivity of the second elongated member.
13. The busbar according to claim 1, wherein the busbar is applied to a battery module, the battery

- module comprises the battery cells having poles, and wherein a first through hole configured to be aligned with a corresponding one of the poles is defined on the connecting part.
- 14.** A battery module, comprising a busbar, wherein the busbar comprising: a bar part; a plurality of conductors arranged at intervals, each of the plurality of conductors comprising a fuse part and a conducting part, wherein the fuse part has a melting point lower than a melting point of the bar part and lower than a melting point of the conducting part, two ends of the fuse part are respectively connected to the conducting part and the bar part, and the conducting part is configured to be connected to a positive electrode or a negative electrode of a battery cell.
- 15.** The battery module according to claim 14, wherein the fuse part extends from the bar part toward the conducting part, and an extending direction of the fuse part forms an acute angle or an obtuse angle with an extending direction of the bar part.
- 16.** The battery module according to claim 14, wherein a cross section of the fuse part perpendicular to a current flowing direction through the fuse part is smaller than a cross section of the conducting part perpendicular to a current flowing direction through the conducting part and is smaller than a cross section of the bar part perpendicular to the current flowing direction through the bar part.
- 17.** The battery module according to claim 16, wherein a fuse hole is defined on the fuse part.
- 18.** The battery module according to claim 16, wherein a fuse slot is defined on the fuse part.
- 19.** The battery module according to claim 17, wherein the fuse hole has an elongated shape, and an extension direction of a wall of the fuse hole is consistent with an extension direction of the fuse part.
- 20.** The battery module according to claim 18, wherein the fuse slot has an elongated shape, and an extension direction of a wall of the fuse slot is consistent with an extension direction of the fuse part.
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